

ELI — Installation

ELI v2.5.21

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Installation — one-click setup, cross-platform (2026-09-11)

> **Updated for v2.4.84 (September 2026).** One hardware-aware GUI wizard for every supported OS. Primary install: GitHub Releases (`ELI-Setup-*.exe`, Linux AppImage, macOS dmg, portable → `./ELI_Setup.sh`). Core wizard stages hard-fail until nomic + voice + chat model are present. Windows desktop = ELI only (GUI singleton). **Home** tab ≠ Home Assistant. See `first time Installation/INSTALLATION_GUIDE.md`.

Runtime OS tools (installed best-effort by `install.sh` / `install.ps1`)

These are **not** Python packages — ELI installs them when a package manager is available, or offers the exact command after a failed action:

Tool	Purpose	Linux	macOS	Windows
`mpv` + `yt-dlp`	YouTube visible playback	apt/dnf/pacman + pip	brew + pip	winget mpv + pip
`tesseract`	Screen OCR	apt/dnf/pacman	brew	winget
`xdotool` / `ydotool`	Mouse/keyboard (X11 / Wayland)	apt/dnf/pacman	—	PyAutoGUI (pip)
`grim`/`slurp`	Wayland screenshots	apt/dnf/pacman	—	native

Tool	Purpose	Linux	macOS	Windows
`playerctl`	Media transport (MPRIS)	apt/dnf/pacman	brew	media keys
`ffmpeg`	Media/Whisper	apt/dnf/pacman	brew	winget

Wayland: `ydotool` needs the `ydotoold` daemon (uinput). `xdotool` only reaches XWayland clients.

See `docs/CROSS_PLATFORM.md` for the full per-OS desktop matrix.

One command per platform sets up Python deps, the **full SQLite architecture** (blank slate — schema only, no personal data), the **nomic embedder**, and the default voice.

Platform	Command	GPU
Linux	`bash install.sh --yes` (`--cpu-only` on Iris Xe / ≤8 GB)	CUDA / CPU
macOS	`bash install.sh --yes`	Metal (no CUDA)
Windows	`ELI-Setup-*.exe` (frozen) or `install.bat` (source)	CUDA pack optional
Android	`bash scripts/install_android.sh`	CPU only (headless)
Portable tarball	`./ELI_Setup.sh` then `./RUN_ELI.sh`	**hardware_policy** (CPU on Iris Xe / ≤8 GB; GPU on NVIDIA/AMD/Arc)

Which download should I get? (GitHub Releases)

Every release ships the platform assets below plus `SHA256SUMS.txt`.

Download	Size	Platform	What's inside	First run needs internet for	Best for
`ELI-Setup-<v>.exe`	~1.2 GB	Windows	Frozen Inno Setup → `%LOCALAPPDATA%\Programs\ELI`	chat model	Most Windows users
`ELI_v2-<v>-windows-x64.zip`	~1.3 GB	Windows	`ELI\ELI.exe` (no Setup.bat / install.ps1)	chat model	Unzip-and-run
`ELI_v2-<v>-macos-arm64.dmg`	~1.4 GB	macOS (Apple Silicon)	App bundle	chat model	Mac users
`ELI_v2-<v>-x86_64.AppImage`	~1.2 GB	Linux, glibc 2.39 or newer	Double-click app	chat model	Most Linux users
`ELI_v2-<v>-linux-portable.tar.gz`	~1.2 GB	Linux, any glibc	Source + `ELI_Setup.sh` (venv)	Python deps + chat model	Older systems / editable / CPU laptops
`SHA256SUMS.txt`	tiny	—	Checksums	—	verifying your download

In plain terms: - Just want it to work → **ELI-Setup-*.exe **(Windows),** .dmg **(macOS),** AppImage **(Linux). - Want the source tree →** linux-portable.tar.gz **→ `./ELI_Setup.sh` only (never `scripts/install_eli.sh`). -** Linux: the AppImage needs glibc 2.39 or newer** (`ldd --version`). That is

Ubuntu 24.04+, Debian 13+, Fedora 40+, Linux Mint 22+, Pop!_OS 24.04+, RHEL / AlmaLinux / Rocky 10+, openSUSE Leap 16+ and Tumbleweed, Arch. On Ubuntu 22.04, Debian 12, Mint 21 or RHEL 9 it does not start and says so; use **linux-portable.tar.gz** there, which installs from source against the system's own libraries. - Frozen builds bundle **nomic** + **Piper voice weights**; Windows TTS still needs a Piper **CLI** on PATH or under `tts_piper/` (weights alone are not enough). - **No download bundles a chat model.** First launch / wizard fetches one. - Windows: desktop shortcut is **ELI only** (Server is Start Menu). If two windows open, close one — v2.4.72+ refuses a second GUI instance.

> **Model size vs. your GPU matters more than which download you pick.** Choose a model that fits your VRAM — e.g. **Qwen3-8B / Qwen2.5-7B** on an 8 GB GPU. Very large models (30B+) run mostly on CPU and will be slow, and the local **web server** may appear to hang on the first message while such a model loads.

Linux — Arch and other lean distros

The **AppImage** is the easiest path on any distro: it bundles its own **Python 3.12** (so whichever Python the distro ships is irrelevant) and, since **v2.1.21**, every Qt xcb library it needs — so it launches out of the box with no extra packages. Download and run it **directly**:

```
U=https://github.com/ShadowESC95/ELI_v2.0/releases/download/v2.4.84
wget "$U/ELI_v2-2.4.84-x86_64.AppImage"
chmod +x ELI_v2-2.4.84-x86_64.AppImage
./ELI_v2-2.4.84-x86_64.AppImage
```

Two fixes worth knowing, both resolved in current builds and verified on a clean Arch VM:

- **"attempt to write a readonly database" (portable build).** If ELI is extracted onto an **NTFS / exFAT / FAT or network** filesystem — e.g. a `~/Downloads` on a dual-boot data drive — SQLite's WAL journal isn't supported there and database init fails (only the `user.*` stores; the others in the same folder succeed, which is the tell-tale). **v2.1.19+** detects a WAL-hostile filesystem and falls back to a rollback journal automatically. On older builds, extract onto an **ext4/btrfs** path instead. Check a mount with `findmnt -T . -o TARGET,FSTYPE`.
- **GUI won't open — "libxcb-cursor0 … xcb platform plugin".** Qt 6.5+ needs the whole **xcb-util family**, which a minimal desktop doesn't ship (Qt's error is misleading — the real missing library is often `libxcb-icccm.so.4`, not the cursor lib). **v2.1.21+** bundles all of them in the AppImage. On an older build or a **source** install, add them:
  - Arch: `sudo pacman -S xcb-util-cursor xcb-util-wm xcb-util-image xcb-util-keysyms xcb-util-renderutil xcb-util`
  - Debian/Ubuntu: `sudo apt install libxcb-cursor0 libxcb-icccm4 libxcb-image0 libxcb-keysyms1 libxcb-render-util0 libxcb-util1 libxcb-shape0`

Run the AppImage **directly** (as above) rather than with `--appimage-extract-and-run`, which unpacks ~4 GB into `/tmp` and can fail with `libz.so.1: file too short` on a small `tmpfs`. Launching over **SSH** into a running desktop session? Prefix with `export DISPLAY=:0`. And paste long download URLs as a **single line** — a wrapped URL 404s and its tail runs as a stray command.

Desktop / applications-menu icons

On its **first launch** the Linux AppImage offers to add **ELI**, **ELI Server** and **Uninstall** to your applications menu (with the app icon), pointing the launchers at the `.AppImage` file. This works when you run the **`.AppImage` directly** *or* via `--appimage-extract-and-run` — both set the `APPIMAGE` path the launcher needs. Force it any time with:

```
./ELI_v2-2.4.84-x86_64.AppImage --integrate      # add/refresh menu entries
./ELI_v2-2.4.84-x86_64.AppImage --uninstall       # remove them
```

Running the **manually extracted `./squashfs-root/AppRun`** does *not* create menu icons — there's no `.AppImage` file for the launcher to point at, so it's skipped (with a one-line note on stderr explaining why). Use the real `.AppImage` for menu integration. On Ubuntu 24.04 a normal double-click/FUSE run

needs `libfuse2` (`sudo apt install libfuse2t64`). The **portable tarball** and **source** installs create the same menu entries via `scripts/install_desktop_apps.sh`; on **Windows** the Setup.exe adds Start-menu shortcuts, and on **macOS** the app is the `.app` bundle you drag to Applications.

CUDA toolkit option (`--install-cuda` / `/cuda`)

For non-technical users with an NVIDIA GPU but no toolkit. Best-effort, never fatal: - **Linux:** tries no-sudo `pip nvidia-cuda-nvcc-cu12` (exposes nvcc via `CUDACXX`), then the system package manager (`apt`/`dnf`/`pacman`, if sudo is available), then prints the manual step — then source-rebuilds llama-cpp with `-DGGML_CUDA=on`. - **Windows:** `winget install Nvidia.CUDA`, then rebuilds llama-cpp. - **macOS/Android:** N/A (Metal / CPU). The default install already uses prebuilt CUDA wheels (no toolkit needed); the option only matters when those don't match the user's CUDA or a source build is required.

What `bash install.sh` does

1. Picks the Python: of those installed at or above the floor `pyproject.toml` declares, the one the most locked packages have ready-made builds for on this machine (asked of PyPI, kept a month; offline, the release's own wheels decide, else the newest). `PYTHON=...` overrides. Detects the OS; checks for what the install builds with (a compiler, Python's headers, PortAudio's, and on Debian/Ubuntu `python3-venv`) and for the system libraries the window loads (GLib, `libGL`, `libEGL`, the xcb helpers), adds what is missing when that takes no password and otherwise prints the one `sudo` command; creates `.venv`; upgrades pip/setuptools/wheel. The `.deb` declares the same packages as dependencies.
2. Installs **PyTorch**: for NVIDIA the newest CUDA build PyTorch publishes for this Python that the driver supports, for AMD the newest ROCm build, for Intel Arc the XPU build (all read from PyTorch's own list, `eli_env.py torch-index`), then checks it runs on the GPU and otherwise takes the oldest one on offer, which keeps older cards; CPU when none runs. macOS uses MPS. PyTorch's outcome never changes how the inference engine is built.
3. Installs **llama-cpp-python** with GPU acceleration (CUDA wheel index / Metal / CPU) — then **verifies `llama_supports_gpu_offload()`** and, if it landed CPU-only, prints the exact CUDA-rebuild command (closes the silent-CPU-wheel trap).
4. Installs the ELI package (`[full]`) + all remaining dependencies from the **frozen `requirements.lock.txt`** (exact known-good versions; reproducible).
5. Seeds `config/settings.json` from the template (offline-by-default, wizard on) — never overwrites an existing config.
6. Runs **`python -m eli.core.init_data`** — creates **every** SQLite store/table (`user.sqlite3`, `system_index.sqlite3`, `coding_memory.sqlite3`, `agent.sqlite3`) with **zero personal memories/profile/history** (blank slate). System inventory (installed apps, `$PATH` binaries) is scanned once so "open Firefox" works — that is machine environment data, not user memories.
7. Verifies, from outside the source folder, that ELI and every package it cannot run without are installed, that the GUI toolkit really loads, and the `eli` console script. An install that ends with errors exits non-zero.
8. Fetches **`nomic-embed-text-v1.5.Q4_K_M.gguf`** → `models/embeddings/` (~80 MiB, required for memory/RAG) via `python -m eli.core.model_download --aux`.
9. Fetches **voice weights**: Piper `en_US-amy-medium` + faster-whisper `small.en` via `python -m eli.runtime.voice_assets` (idempotent; skipped if already present).
10. Optionally downloads a **chat GGUF** (wizard / `--model=` / `--auto`).
11. Prints how to launch + how to add more models.

Three local asset layers (not in git)

Asset	Path	Size (typical)	When fetched
Embedder (required)	`models/embeddings/nomic-embed-text-v1.5.Q4_K_M.gguf`	~80 MiB	`install.sh` / wizard / `--aux`
Chat model (pick one+)	`models/*.gguf`	0.6–8+ GiB	Wizard / `model_download` / asset pack
Voice (default amy)	`models/tts/piper/` + `tts_piper/piper/`	~60 MiB Piper + ~464 MiB whisper	`voice_assets` / asset pack

GitHub Release tag **`local-assets-v2.1`** mirrors embedder + starter chat GGUFs + cleared Piper voices. Restore: `./RUN_ELI.sh --with-github-assets`. Voices excluded from auto-restore: ryan, lessac, cori — see `models/MODEL_LICENSES.md`.

Flags

- `--cpu-only` — no CUDA (CPU torch + CPU llama-cpp).
- `--latest` — use `requirements.txt` version ranges instead of the frozen lock.
- `--skip-torch` — leave an existing torch in place.
- `--no-model` — skip chat-model offer; embedder still fetched unless you disable network.

Files

- `install.sh` (Linux/macOS), `install.bat` / `install.ps1` (Windows).
- `requirements.lock.txt` — frozen exact versions, with markers for what only some Pythons or systems need (excludes torch and llama-cpp, which the installers fetch for the machine's GPU).
- `pyproject.toml` — package metadata + `[project.scripts]` (`eli` → `eli.gui.app:main`).

Launch

`./eli.sh` or `source .venv/bin/activate && eli`. First run shows the **setup wizard** if no chat GGUF is present. Wizard also verifies embedder + voice and can fetch them.

Chat model: `python -m eli.core.model_download --auto` (or `--list`, or a named model). ELI stays offline by default; downloads are deliberate one-time actions.

Training dependencies now ship

`peft` and `datasets` were previously only in `requirements.lock.txt`, so a plain `pip install -r requirements.txt` produced an install whose Training tab could never leave preflight — it reported the packages missing and stopped. Both are now in `requirements.txt`, alongside:

```
peft==0.19.1
datasets==4.8.5
bitsandbytes>=0.43; platform_system != "Darwin"
```

`bitsandbytes` enables 4-bit (QLoRA) training, which lets a card too small for the full-precision weights train an adapter anyway. It is **optional everywhere** and the trainer falls back to fp16 without it: there is no macOS wheel, and AMD needs a separate ROCm build. `install.sh` is unchanged — it already installs from the frozen lock, which carried both packages.

Optional scanners for the plugin marketplace

Neither is required, and their absence is reported rather than silently ignored:

- **ClamAV** (`clamscan` / `clamdscan` on PATH) — full antivirus over plugin sources.

- **yara-python** plus a ruleset at `<config dir>/plugin_yara_rules.yar`.

Without them the marketplace still runs its nine built-in engines and states that coverage was partial.

Optional runtimes for MCP servers

MCP servers are separate programs. `npx` (Node.js) and `uvx` (uv) cover almost all published servers. ELI checks for the required runtime **before** writing anything to its MCP config and names the install command for your platform if it is missing.